

Notes on Multinomial Model

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1 General Multinomial Model

$$y_i \in \{1, \dots, m\}$$

Define m dummy variables

$$y_{ij} = \begin{cases} 1 & y_i = j \\ 0 & y_i \neq j \end{cases} \quad \text{for } j \in \{1, \dots, m\}$$

Model p_{ij}

$$p_{ij} := \Pr(y_i = j | \mathbf{x}_i) = F_j(\mathbf{x}_i, \boldsymbol{\beta}) \quad \text{for } j \in \{1, \dots, m\}$$

where $\sum_{j=1}^m p_{ij} = 1$

1.1 ML Estimation

Log likelihood function

$$f(y_i) = p_{i1}^{y_{i1}} \cdots p_{im}^{y_{im}} = \prod_{j=1}^m p_{ij}^{y_{ij}}$$

$$\begin{aligned} \ln[L_N(\boldsymbol{\beta})] &= \ln\left[\prod_{i=1}^N f(y_i)\right] && \text{assume independence} \\ &= \sum_{i=1}^N \ln[f(y_i)] \\ &= \sum_{i=1}^N \ln\left[\prod_{j=1}^m F_j(\mathbf{x}_i, \boldsymbol{\beta})^{y_{ij}}\right] \\ &= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln[F_j(\mathbf{x}_i, \boldsymbol{\beta})] \end{aligned}$$

$$\begin{aligned}
\frac{\partial \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} &= \frac{\partial \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln[F_j(\mathbf{x}_i, \boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} \\
&= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \frac{\partial \ln[F_j(\mathbf{x}_i, \boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} \\
&= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \frac{1}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}
\end{aligned}$$

FOC

$$\sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \Big|_{\hat{\boldsymbol{\beta}}} = \mathbf{0}$$

1.2 Consistency of MLE

It requires the correct specification of $F_j(\mathbf{x}_i, \boldsymbol{\beta})$

1.3 Asymptotic Distribution of MLE

$$\hat{\boldsymbol{\beta}} \sim_a N(\boldsymbol{\beta}_0, [\mathbf{I}(\boldsymbol{\beta}_0)]^{-1}) = N(\boldsymbol{\beta}_0, [\sum_{i=1}^N \sum_{j=1}^m \{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-1} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} - \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0}\}]^{-1})$$

$$\begin{aligned}
\mathbf{I}(\boldsymbol{\beta}_0) &:= -\mathbb{E}\left[\frac{\partial^2 \ln L_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] = \mathbb{E}\left[\frac{\partial \ln L_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \cdot \frac{\partial \ln L_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] \\
&= -\mathbb{E}\left[\frac{\partial \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] \\
&= -\sum_{i=1}^N \sum_{j=1}^m \mathbb{E}\left[\frac{\partial \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] \\
&= -\sum_{i=1}^N \sum_{j=1}^m \mathbb{E}\left[y_{ij} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-1}}{\partial \boldsymbol{\beta}'} + \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big| \mathbf{x}_i\right]_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \mathbb{E}\left[-y_{ij} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} + \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big| \mathbf{x}_i\right]_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -\mathbb{E}[y_{ij} | \mathbf{x}_i] \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} + \frac{\mathbb{E}[y_{ij} | \mathbf{x}_i]}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right\} \Big|_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -F_j(\mathbf{x}_i, \boldsymbol{\beta}_0) \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} + \frac{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right\} \Big|_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -F_j(\mathbf{x}_i, \boldsymbol{\beta}_0) \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \Big|_{\boldsymbol{\beta}_0} F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} + \frac{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)}{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \right\} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-1} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} + \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \right\} \\
&= \sum_{i=1}^N \sum_{j=1}^m \left\{ F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-1} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} - \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \right\}
\end{aligned}$$

1.4 McFadden's Pseudo- R^2

$$\begin{aligned}
R_{multi}^2 &= 1 - \frac{\ln L_{fit}}{\ln L_0} \\
&= \frac{\ln L_0 - \ln L_{fit}}{\ln L_0} \\
&= \frac{\ln L_0 - \ln L_{fit}}{\ln L_0 - 0} \\
&= \frac{\ln L_0 - \ln L_{fit}}{\ln L_0 - \ln L_{max}} \\
&= \frac{\ln L_{fit} - \ln L_0}{\ln L_{max} - \ln L_0}
\end{aligned}$$

$$L_{max} = \prod_{i=1}^N \prod_{j=1}^m p_{ij}^{y_{ij}} = \prod_{i=1}^N 0^0 \dots 1^1 \dots 0^0 = \prod_{i=1}^N 1 \dots 1 \dots 1 = \prod_{i=1}^N 1 = 1$$

1.5 Special Case: Conditional Logit Model

if $F_j(\mathbf{x}_i, \boldsymbol{\beta}) = F_{softmax}(\mathbf{x}'_{ij}\boldsymbol{\beta})$

$$p_{ij} := Pr(y_i = j | \mathbf{x}_i) = F_{softmax}(\mathbf{x}'_{ij}\boldsymbol{\beta}) = \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \quad \text{for } j \in \{1, \dots, m\}$$

1.5.1 Gradient Vector

$$\begin{aligned}
\frac{\partial p_{ij}}{\partial \boldsymbol{\beta}} &= \frac{\partial \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})}}{\partial \boldsymbol{\beta}} \\
&= \frac{\partial \exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^{-1} + \frac{\partial \left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^{-1}}{\partial \boldsymbol{\beta}} \exp(\mathbf{x}'_{ij}\boldsymbol{\beta}) \\
&= \exp(\mathbf{x}'_{ij}\boldsymbol{\beta}) \mathbf{x}_{ij} \left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^{-1} - \left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^{-2} \left[\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \mathbf{x}_{il} \right] \exp(\mathbf{x}'_{ij}\boldsymbol{\beta}) \\
&= \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \mathbf{x}_{ij} - \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^2} \left[\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \mathbf{x}_{il} \right] \\
&= \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \mathbf{x}_{ij} - \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \left[\sum_{l=1}^m \frac{\exp(\mathbf{x}'_{il}\boldsymbol{\beta})}{\sum_{k=1}^m \exp(\mathbf{x}'_{ik}\boldsymbol{\beta})} \mathbf{x}_{il} \right] \\
&= p_{ij} \mathbf{x}_{ij} - p_{ij} \left[\sum_{l=1}^m p_{il} \mathbf{x}_{il} \right] \\
&= p_{ij} \mathbf{x}_{ij} - p_{ij} \bar{\mathbf{x}}_i \\
&= p_{ij} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)
\end{aligned}$$

$\bar{\mathbf{x}}_i = \sum_{l=1}^m p_{il} \mathbf{x}_{il}$

$$\frac{\partial \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} = \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} \frac{\partial p_{ij}}{\partial \boldsymbol{\beta}} = \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} p_{ij} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i) = \sum_{i=1}^N \sum_{j=1}^m y_{ij} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)$$

1.5.2 Asymptotic Distribution

$$\begin{aligned}
\frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} &= \frac{\sum_{i=1}^N \sum_{j=1}^m \partial y_{ij}(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)}{\partial \boldsymbol{\beta}'} = \sum_{i=1}^N \sum_{j=1}^m \frac{\partial y_{ij}(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)}{\partial \boldsymbol{\beta}'} \\
&= - \sum_{i=1}^N \sum_{j=1}^m \frac{\partial y_{ij} \bar{\mathbf{x}}_i}{\partial \boldsymbol{\beta}'} \\
&= - \sum_{i=1}^N \sum_{j=1}^m \frac{\partial y_{ij} \sum_{l=1}^m p_{il} \mathbf{x}_{il}}{\partial \boldsymbol{\beta}'} \\
&= - \sum_{i=1}^N \sum_{j=1}^m y_{ij} \sum_{l=1}^m \frac{\partial p_{il}}{\partial \boldsymbol{\beta}'} \mathbf{x}_{il} \\
&= - \sum_{i=1}^N \sum_{j=1}^m y_{ij} \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i)' \mathbf{x}_{il} \\
&= - \sum_{i=1}^N \left[\sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) \mathbf{x}_{il}' \right] \sum_{j=1}^m y_{ij} \\
&= - \sum_{i=1}^N \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) \mathbf{x}_{il}' \quad \sum_j y_{ij} = 1 \\
&= - \sum_{i=1}^N \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) (\mathbf{x}_{il} - \bar{\mathbf{x}}_i)'
\end{aligned}$$

The last line is due to $\sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = \sum_{l=1}^m (p_{il} \mathbf{x}_{il} - p_{il} \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = (\sum_{l=1}^m p_{il} \mathbf{x}_{il} - [\sum_{l=1}^m p_{il}] \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = (\bar{\mathbf{x}}_i - 1 \cdot \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = \mathbf{0}$

Thus, the asymptotic variance is $[\mathbf{I}(\boldsymbol{\beta}_0)]^{-1} := [-\mathbb{E}(\frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} | \mathbf{x}_i)]^{-1} = [\sum_{i=1}^N \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) (\mathbf{x}_{il} - \bar{\mathbf{x}}_i)']^{-1}$

1.6 Special Case: Multinomial Logit Model

if $F_j(\mathbf{x}_i, \boldsymbol{\beta}) = F_{\text{softmax}}(\mathbf{x}_i' \boldsymbol{\beta}_j)$

$$p_{ij} := \Pr(y_i = j | \mathbf{x}_i) = F_{\text{softmax}}(\mathbf{x}_i' \boldsymbol{\beta}_j) = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta}_j)}{\sum_{l=1}^m \exp(\mathbf{x}_i' \boldsymbol{\beta}_l)} \quad \text{for } j \in \{1, \dots, m\}$$

1.6.1 Gradient Vector

$$\begin{aligned}
\frac{\partial p_{ij}}{\partial \beta_j} &= \frac{\partial \frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)}}{\partial \beta_j} \\
&= \frac{\partial \exp(\mathbf{x}'_i \beta_j)}{\partial \beta_j} \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-1} + \frac{\partial (\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l))^{-1}}{\partial \beta_j} \exp(\mathbf{x}'_i \beta_j) \\
&= \exp(\mathbf{x}'_i \beta_j) \mathbf{x}_i \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-1} - \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \left[\sum_{l=1}^m \frac{\partial \exp(\mathbf{x}'_i \beta_l)}{\partial \beta_j} \right] \exp(\mathbf{x}'_i \beta_j) \\
&= \exp(\mathbf{x}'_i \beta_j) \mathbf{x}_i \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-1} - \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \exp(\mathbf{x}'_i \beta_j) \mathbf{x}_i \exp(\mathbf{x}'_i \beta_j) \\
&= \frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \mathbf{x}_i - \left(\frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \right)^2 \mathbf{x}_i \\
&= p_{ij} \mathbf{x}_i - p_{ij}^2 \mathbf{x}_i
\end{aligned}$$

$$\begin{aligned}
\frac{\partial p_{ij}}{\partial \beta_k} &= \frac{\partial \frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)}}{\partial \beta_k} \\
&= \exp(\mathbf{x}'_i \beta_j) \frac{\partial (\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l))^{-1}}{\partial \beta_k} \\
&= -\exp(\mathbf{x}'_i \beta_j) \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \sum_{l=1}^m \frac{\partial \exp(\mathbf{x}'_i \beta_l)}{\partial \beta_k} \\
&= -\exp(\mathbf{x}'_i \beta_j) \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \exp(\mathbf{x}'_i \beta_k) \mathbf{x}_i \\
&= -\frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \frac{\exp(\mathbf{x}'_i \beta_k)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \mathbf{x}_i \\
&= -p_{ij} p_{ik} \mathbf{x}_i
\end{aligned}$$

Thus,

$$\frac{\partial p_{ij}}{\partial \beta_l} = 1(l = j)p_{ij}\mathbf{x}_i - p_{ij}p_{il}\mathbf{x}_i = p_{ij}(1(l = j) - p_{il})\mathbf{x}_i$$

$$\begin{aligned} \frac{\partial \ln[L_N(\boldsymbol{\beta})]}{\partial \beta_l} &= \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} \frac{\partial p_{ij}}{\partial \beta_l} \\ &= \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} p_{ij}(1(l = j) - p_{il})\mathbf{x}_i \\ &= \sum_{i=1}^N \mathbf{x}_i \sum_{j=1}^m y_{ij}(1(l = j) - p_{il}) \\ &= \sum_{i=1}^N \mathbf{x}_i [-p_{il} \sum_{j \neq l}^m y_{ij} + y_{il}(1 - p_{il})] \\ &= \sum_{i=1}^N \mathbf{x}_i [-p_{il} \sum_{j \neq l}^m y_{ij} + y_{il} - y_{il}p_{il}] \\ &= \sum_{i=1}^N \mathbf{x}_i [-p_{il} \sum_{j=1}^m y_{ij} + y_{il}] \\ &= \sum_{i=1}^N \mathbf{x}_i [y_{il} - p_{il}] \end{aligned} \quad \text{As } \sum_{j=1}^m y_{ij} = 1$$

1.6.2 Asymptotic Distribution

$$\begin{aligned} \frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \beta_j \partial \beta'_k} &= \frac{\partial \sum_{i=1}^N \mathbf{x}_i [y_{ij} - p_{ij}]}{\partial \beta'_k} \\ &= - \sum_{i=1}^N \frac{\partial p_{ij}}{\partial \beta'_k} \mathbf{x}_i \\ &= - \sum_{i=1}^N p_{ij}(1(k = j) - p_{ik})\mathbf{x}_i \mathbf{x}'_i \end{aligned}$$

Thus, the asymptotic variance is $[\mathbf{I}(\boldsymbol{\beta}_0)]^{-1} := [-\mathbb{E}(\frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \beta_j \partial \beta'_k} | \mathbf{x}_i)]^{-1} = [\sum_{i=1}^N p_{ij}(1(k = j) - p_{ik})\mathbf{x}_i \mathbf{x}'_i]^{-1}$

1.7 Special Case: Universal Logit Model

if $F_j(\mathbf{x}_i, \boldsymbol{\beta}) = F_{softmax}(\ln[V_{ij}])$

$$p_{ij} := \Pr(y_i = j | \mathbf{x}_i) = F_{softmax}(\ln[V_{ij}]) = \frac{\exp(\ln[V_{ij}])}{\sum_{l=1}^m \exp(\ln[V_{il}])} = \frac{V_{ij}}{\sum_{l=1}^m V_{il}} \quad \text{for } j \in \{1, \dots, m\}$$

2 Additive Random Utility Model

Suppress i ,

$$U_j = V_j + \varepsilon_j \quad j \in \{1, \dots, m\}$$

$$\begin{aligned} p_j &:= Pr(y = j | \mathbf{x}) \\ &= Pr(U_j > U_k \ \forall k \neq j | \mathbf{x}) \\ &= Pr(V_j + \varepsilon_j > V_k + \varepsilon_k \ \forall k \neq j | \mathbf{x}) \\ &= Pr(V_j - V_k > \varepsilon_k - \varepsilon_j \ \forall k \neq j | \mathbf{x}) \\ &= Pr(\varepsilon_k - \varepsilon_j < V_j - V_k \ \forall k \neq j | \mathbf{x}) \\ &= Pr(\tilde{\varepsilon}_{kj} < -\tilde{V}_{kj} \ \forall k \neq j | \mathbf{x}) \\ &= F_j(-\tilde{V}_{kj} \ \forall k \neq j) \\ &= \int_{-\infty}^{-\tilde{V}_{1j}} \dots \int_{-\infty}^{-\tilde{V}_{j-1,j}} \int_{-\infty}^{-\tilde{V}_{j+1,j}} \dots \int_{-\infty}^{-\tilde{V}_{mj,j}} f(\tilde{\varepsilon}_{1j}, \dots, \tilde{\varepsilon}_{j-1,j}, \tilde{\varepsilon}_{j+1,j}, \dots, \tilde{\varepsilon}_{mj}) \partial \tilde{\varepsilon}_{1j} \dots \partial \tilde{\varepsilon}_{j-1,j} \partial \tilde{\varepsilon}_{j+1,j} \dots \partial \tilde{\varepsilon}_{mj} \end{aligned}$$

2.1 Special Case: Conditional Logit Model, Multinomial Logit Model

If $\varepsilon_k \ \forall k$ are independent and all follow Type 1 Extreme Value Distribution (or log Weibull Distribution). It can be shown $p_j := Pr(y = j | \mathbf{x}) = \frac{\exp(V_j)}{\sum_{l=1}^m \exp(V_l)}$ by direct integration method. If $V_{ij} = \mathbf{x}'_{ij} \boldsymbol{\beta}$, it is Conditional Logit Model. If $V_{ij} = \mathbf{x}'_i \boldsymbol{\beta}_j$, it is Multinomial Logit Model.

2.2 Special Case: Generalized Extreme Value Model (McFadden, 1978)

Nested Logit Model is a special case of GEV model. Conditional Logit and Multinomial Logit Models are special case of Nested Logit Model.

If $\boldsymbol{\varepsilon}$ follows (multivariate) GEV distribution i.e., $F(\boldsymbol{\varepsilon}) = F(\varepsilon_1, \dots, \varepsilon_m) = \exp[-G(e^{-\varepsilon_1}, \dots, e^{-\varepsilon_m})]$. If $G(\cdot)$ satisfies certain conditions, then $p_j := Pr(y = j | \mathbf{x}) = e^{V_j} \frac{G_j(e^{-V_1}, \dots, e^{-V_m})}{G(e^{-V_1}, \dots, e^{-V_m})}$ where G_j is derivative of the j argument.

2.2.1 Nested Logit Model

$$F(\varepsilon_{11}, \dots, \varepsilon_{1K_1}; \dots; \varepsilon_{J1}, \dots, \varepsilon_{JK_J}) = \exp[-G(e^{-\varepsilon_{11}}, \dots, e^{-\varepsilon_{1K_1}}; \dots; e^{-\varepsilon_{J1}}, \dots, e^{-\varepsilon_{JK_J}})]$$

$$p_{jk} := Pr(y_{jk} = 1 | \mathbf{x}) = e^{V_{jk}} \frac{G_{jk}(e^{-V_{11}}, \dots, e^{-V_{1K_1}}; \dots; e^{-V_{J1}}, \dots, e^{-V_{JK_J}})}{G(e^{-V_{11}}, \dots, e^{-V_{1K_1}}; \dots; e^{-V_{J1}}, \dots, e^{-V_{JK_J}})}$$

Let $Y_{jk} = e^{V_{jk}}$ and specify

$$G(\mathbf{Y}) = G(Y_{11}, \dots, Y_{1K_1}; \dots; Y_{J1}, \dots, Y_{JK_J}) = \sum_{j=1}^J a_j \left(\sum_{k=1}^{K_j} Y_{jk}^{1/\rho_j} \right)^{\rho_j}$$

We have

$$\begin{aligned}
G_{j'k'} &= \frac{\partial G(\mathbf{Y})}{\partial Y_{j'k'}} \\
&= \frac{\partial \sum_{j=1}^J a_j (\sum_{k=1}^{K_j} Y_{jk}^{1/\rho_j})^{\rho_j}}{\partial Y_{j'k'}} \\
&= a_{j'} \frac{\partial (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}}}{\partial Y_{j'k'}} \\
&= a_{j'} \rho_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} \sum_{k=1}^{K_{j'}} \frac{\partial Y_{j'k}^{1/\rho_{j'}}}{\partial Y_{j'k'}} \\
&= a_{j'} \rho_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} \frac{\partial Y_{j'k'}^{1/\rho_{j'}}}{\partial Y_{j'k'}} \\
&= a_{j'} \rho_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} (1/\rho_{j'}) Y_{j'k'}^{1/\rho_{j'}-1} \\
&= a_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} Y_{j'k'}^{1/\rho_{j'}-1} \\
Y_{j'k'} G_{j'k'} &= a_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} Y_{j'k'}^{1/\rho_{j'}}
\end{aligned}$$

$$\begin{aligned}
p_{jk} &= Y_{jk} \frac{G_{jk}}{G(\mathbf{Y})} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
p_j &:= \sum_{k=1}^{K_j} p_{jk} = \sum_{k=1}^{K_j} \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} \sum_{k=1}^{K_j} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
p_{k|j} &= \frac{p_{jk}}{p_j} = \frac{\frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}}}{\frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}} \\
&= \frac{Y_{jk}^{1/\rho_j}}{\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j}}
\end{aligned}$$

Substitute back

$$\begin{aligned}
(e^{V_{jl}})^{1/\rho_j} &= (e^{\mathbf{z}'_j \boldsymbol{\alpha} + \mathbf{x}'_{jl} \boldsymbol{\beta}_j})^{1/\rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j + \mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}
\end{aligned}$$

$$\begin{aligned}
\sum_{l=1}^{K_j} (e^{V_{jl}})^{1/\rho_j} &= \sum_{l=1}^{K_j} e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \exp(\ln[\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}]) \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \exp(l_j)
\end{aligned}$$

$$\begin{aligned}
(\sum_{l=1}^{K_j} (e^{V_{jl}})^{1/\rho_j})^{\rho_j} &= (\exp(\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j) \exp(l_j))^{\rho_j} \\
&= (\exp(\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j + l_j))^{\rho_j} \\
&= \exp(\mathbf{z}'_j \boldsymbol{\alpha} + \rho_j l_j)
\end{aligned}$$

$$\begin{aligned}
p_j &= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} e^{V_{jl}^{1/\rho_j}})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} e^{V_{ml}^{1/\rho_m}})^{\rho_m}} \\
&= \frac{a_j \exp(\mathbf{z}'_j \boldsymbol{\alpha} + \rho_j l_j)}{\sum_{m=1}^J a_m \exp(\mathbf{z}'_m \boldsymbol{\alpha} + \rho_m l_m)}
\end{aligned}$$

$$\begin{aligned}
p_{k|j} &= \frac{Y_{jk}^{1/\rho_j}}{\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j}} \\
&= \frac{e^{V_{jk}^{1/\rho_j}}}{\sum_{l=1}^{K_j} e^{V_{jl}^{1/\rho_j}}} \\
&= \frac{e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}} \\
&= \frac{e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}} \\
&= \frac{e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}}
\end{aligned}$$

Thus,

$$\begin{aligned}
p_{jk} &= p_{k|j} p_j \\
&= \frac{e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}} \frac{a_j \exp(\mathbf{z}'_j \boldsymbol{\alpha} + \rho_j l_j)}{\sum_{m=1}^J a_m \exp(\mathbf{z}'_m \boldsymbol{\alpha} + \rho_m l_m)}
\end{aligned}$$

where $l_j = \ln[\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl}\boldsymbol{\beta}_j/\rho_j}]$ is log sum

It can be shown that $\rho_j = \sqrt{1 - \text{Corr}[\varepsilon_{jk}, \varepsilon_{jl}]}$. $\rho_j = 1 \iff \text{Corr}[\varepsilon_{jk}, \varepsilon_{jl}] = 0$ i.e., Multinomial Logit Model

2.2.2 FIML Estimation of Nested Logit Model

$$\begin{aligned}
f(\mathbf{y}_i) &= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ijk}^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} (p_{ik|j} p_{ij})^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} p_{ij}^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} \prod_{k=1}^{K_j} p_{ij}^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} \cdot p_{ij}^{\sum_{k=1}^{K_j} y_{ijk}} \\
&= \prod_{j=1}^J p_{ij}^{\sum_{k=1}^{K_j} y_{ijk}} \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} \\
&= \prod_{j=1}^J p_{ij}^{y_{ij}} \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}}
\end{aligned}$$

$$\begin{aligned}
\ln[L(\boldsymbol{\alpha}, \boldsymbol{\beta}_j, \rho_j)] &= \ln\left[\prod_{i=1}^N f(\mathbf{y}_i)\right] && \text{assume independence of i} \\
&= \ln\left[\prod_{i=1}^N \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} p_{ij}^{y_{ijk}}\right] \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} \ln[p_{ik|j}^{y_{ijk}} p_{ij}^{y_{ijk}}] \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} \{y_{ijk} \ln[p_{ik|j}] + y_{ijk} \ln[p_{ij}]\} \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln[p_{ik|j}] + \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln[p_{ij}] \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln[p_{ik|j}] + \sum_{i=1}^N \sum_{j=1}^J \ln[p_{ij}] \underbrace{\sum_{k=1}^{K_j} y_{ijk}}_{y_{ij}} \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln\left[\frac{e^{\mathbf{x}'_{ijk}\boldsymbol{\beta}_j/\rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{ijl}\boldsymbol{\beta}_j/\rho_j}}\right] + \sum_{i=1}^N \sum_{j=1}^J y_{ij} \ln\left[\frac{\exp(\mathbf{z}'_{ij}\boldsymbol{\alpha} + \rho_j l_{ij})}{\sum_{m=1}^J \exp(\mathbf{z}'_{im}\boldsymbol{\alpha} + \rho_m l_{im})}\right]
\end{aligned}$$

2.2.3 LIML Estimation of Nested Logit Model

It is less efficient compared to FIML

First Stage

Estimate β_j/ρ_j with maximizing log likelihood $\sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln \left[\frac{e^{\mathbf{x}'_{ijk} \beta_j / \rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{ijl} \beta_j / \rho_j}} \right]$

Second Stage

Estimate α and ρ_j with maximizing log likelihood $\sum_{i=1}^N \sum_{j=1}^J y_{ij} \ln \left[\frac{\exp(\mathbf{z}'_{ij} \alpha + \rho_j l_{ij})}{\sum_{m=1}^J \exp(\mathbf{z}'_{im} \alpha + \rho_m l_{im})} \right]$

Get β_j by $\beta_j / \rho_j \cdot \rho_j$

2.3 Random Parameters Logit Model

If $V_{ij} = \mathbf{x}'_{ij} \beta_i$ i.e., Conditional Logit Model with heterogeneous beta.

$$U_{ij} = \mathbf{x}'_{ij} \beta_i + \varepsilon_{ij}$$

where

$$\beta_i \sim N(\beta, \Gamma_\beta)$$

It can be written as

$$\begin{aligned} U_{ij} &= \mathbf{x}'_{ij} (\beta + \mathbf{u}_i) + \varepsilon_{ij} & \mathbf{u}_i &\sim N(\mathbf{0}, \Gamma_\beta) \\ &= \mathbf{x}'_{ij} \beta + \underbrace{(\mathbf{x}'_{ij} \mathbf{u}_i + \varepsilon_{ij})}_{v_{ij}} \end{aligned}$$

$Cov(v_{ij}, v_{ik}) = Cov(\mathbf{x}'_{ij} \mathbf{u}_i + \varepsilon_{ij}, \mathbf{x}'_{ik} \mathbf{u}_i + \varepsilon_{ik}) = Cov(\mathbf{x}'_{ij} \mathbf{u}_i, \mathbf{u}'_i \mathbf{x}_{ik}) = \mathbf{x}'_{ij} \Gamma_\beta \mathbf{x}_{ik}$ for $j \neq k$ i.e., Conditional Logit Model with correlated error v_{ij} across alternative.

2.3.1 Maximum Simulated Estimation

$$\begin{aligned} p_{ij} &:= Pr(y_i = j | \mathbf{x}_i) = \int Pr(y_i = j, \beta_i | \mathbf{x}_i, \beta, \Gamma_\beta) d\beta_i \\ &= \int Pr(y_i = j | \beta_i, \mathbf{x}_i, \beta, \Gamma_\beta) \phi(\beta_i | \mathbf{x}_i, \beta, \Gamma_\beta) d\beta_i \\ &= \int Pr(y_i = j | \beta_i, \mathbf{x}_i) \phi(\beta_i | \beta, \Gamma_\beta) d\beta_i \\ &= \int \frac{\exp(\mathbf{x}'_{ij} \beta_i)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i)} \phi(\beta_i | \beta, \Gamma_\beta) d\beta_i \\ &= \mathbb{E} \left[\frac{\exp(\mathbf{x}'_{ij} \beta_i)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i)} \right] \end{aligned}$$

Simulate p_{ij} by drawing β_i from $\phi(\beta_i | \beta, \Gamma_\beta)$ S times

$$\hat{p}_{ij} = S^{-1} \sum_{s=1}^S \frac{\exp(\mathbf{x}'_{ij} \beta_i^s)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i^s)}$$

Thus, the log simulated likelihood function is

$$\begin{aligned} \ln[\hat{L}] &= \ln \left[\prod_{i=1}^N \prod_{j=1}^m \hat{p}_{ij}^{y_{ij}} \right] \\ &= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln[\hat{p}_{ij}] \\ &= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln \left[S^{-1} \sum_{s=1}^S \frac{\exp(\mathbf{x}'_{ij} \beta_i^s)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i^s)} \right] \end{aligned}$$

3 References

Cameron, A. C., & Trivedi, P. K. (2005). *Microeconometrics: Methods and Applications*.